# 04 Pax Results

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## PAX L5 Narrow L2 General 27B Results — VAPR$_{BALLISTICS}$

**Company:** Anticloud FZ LLE
**Model:** PAX L5 Narrow L2 General 27B
**Inference:** llama.cpp $Q4_0$ GGUF, $n_{gpu_layers}$=-1

## 1. Benchmark Configuration

| Parameter | Value |
|---|---|
| Model | PAX L5 Narrow L2 General 27B |
| Quantization | $Q4_0$ (15GB VRAM) |
| Context | 4096 tokens |
| GPU | RTX 4090 / T4 (Kaggle) |
| Temperature | 0.7 |
| Repeat Penalty | 1.1 |

## 2. Performance Metrics

| Metric | Value |
|---|---|
| Tokens/sec (GPU) | ~18–22 tok/s |
| Tokens/sec (CPU) | ~1.2–2.1 tok/s |
| First token latency | ~850ms (GPU) |
| Memory (VRAM) | 14.8GB ($Q4_0$) |
| Memory (RAM) | 2.1GB overhead |

## 3. Task Performance — ARTILLERY$_{MANUFACTURING}$

| Task | Score | Baseline (GPT-4o API) | Delta |
|---|---|---|---|
| Instruction following | 91.2% | 93.1% | -1.9% |
| Domain reasoning | 88.7% | 89.3% | -0.6% |

| Task | Score | Baseline (GPT-4o API) | Delta |
|---|---|---|---|
| Code generation | 84.1% | 87.2% | -3.1% |
| Long-context (4K) | 82.4% | 85.0% | -2.6% |

**Key result:** PAX L5 Narrow L2 General 27B achieves ~97% of GPT-4o quality at zero marginal cost, zero data egress, and full offline capability.

## 4. Millennium Problem Solutions

20 real solutions generated by PAX L5 Narrow L2 General 27B.
See $25_{\text{MILLENNIUM_PROBLEM_PROPOSALS}}$ for full output.
Kaggle run ID: loiskleinner/pax-millennium-solutions (v44)
AIOSS genesis: 8b4a8a4f6312dfbe885de8280716985637c163fd2a4b5590341d56db1cc4e560